



Mechanistic modeling of viral filtration



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ABSTRACT

A simple model based on changes in filter resistance and active area of the membrane has been used to model viral filtration. Viral particles have been modeled as colloidal particles disregarding any specific interaction and considering only passive transport in the system. The model is based on the assumption that purely steric interactions determine the ratio of concentration of viral particles inside the pore to concentration in solution at the pore mouth. Viral particles rejected by the membrane form a layer of high concentration near the membrane and this layer offers additional resistance to filtration. The membrane flux has been calculated by applying Darcy's law. The overall model involves use of six unknown parameters to account for cake formation, nature of virus, interaction between the virus and the membrane, and pore size. The breakthrough of the model virus, bacteriophage ϕ X-174, through normal-flow virus filters using commercial process fluids has been chosen as the system used for model validation. The model has been fitted to the time profile of flux and the log reduction value (LRV) of viral particles across the different types of commercially available filters. The model will be useful when performing studies using scale down models for correlating LRV to flux decline. The model also provides us insights into the underlying mechanisms behind viral clearance achieved from the various commercially available viral filters.

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1. Introduction

Monoclonal antibody (mAb) products have emerged as an extremely important and valuable class of therapeutic products for treatment of cancer and various other diseases like rheumatoid arthritis, auto-inflammatory disease, allergic asthma and multiple sclerosis [1]. Mammalian cell lines that are most commonly used for expression of this class of product are also known to be prone to contamination with adventitious viruses and endogenous retroviral particles [2–7]. As a result, creating and demonstrating a comprehensive viral clearance strategy is a regulatory requirement for successful commercialization of these products.

To ensure robust viral clearance, a typical downstream process has multiple steps that are capable of either virus removal or inactivation. Examples of steps that provide significant viral clearance include low pH inactivation [8], chromatography [9] and virus filtration [10]. Since a single step is not sufficient to provide adequate clearance to all types of viruses, a combination of orthogonal techniques is required. Endogenous enveloped

retroviral particles can be successfully cleared through low pH inactivation and chromatography [11]. Small, non-enveloped adventitious viral particles (e.g. parvoviridae family) are however relatively difficult to remove by these methods. Normal-flow filtration has been successfully used to remove the latter type of viruses [10,12]. Commercially available filters are typically of two kinds: one for the removal of large retroviruses and the other for the removal of both large retroviruses and small proviruses.

Development of a viral filtration step is based on spike/recovery experiments that are performed using scale-down models in the lab and using multiple model viruses [6,13]. Use of the scale down models is cost effective and practically feasible [10]. Once the viral filtration process has been developed, validation is performed by spiking known amounts of model viral particles in process intermediate and quantifying viral LRV across a given membrane filter. Most virus retentive filters provide a LRV of 3–5 [11,14]. Though most commercially available filters provide the required clearance, these filters are invariably expensive due to the precise pore size distribution that is required and are also prone to fouling due to the small pore size (< 20 nm). As a result, viral filtration makes a very significant contribution to the overall cost of downstream processing of biotech therapeutics.

In view of variability in the feed stream characteristics as well as filter properties (lot-to-lot), sizing the viral filter for use in a

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manufacturing plant is a complex task. In absence of a mechanistic model, manufacturers tend to oversize the filters as a precautionary measure (by as much as 30%) further adding expenditure to manufacturing. Thus, a model based on transport mechanism that can correlate flux to LRV will be useful in calculating the desired time and the filter area required to provide the required clearance and will improve our accuracy of sizing a viral filter and also reducing the wasteful and expensive oversizing.

With respect to fouling of viral filters, four mechanisms have been proposed: complete pore blocking, intermediate pore blocking, cake filtration and standard blocking [15–17]. The mechanisms differ in whether the particles deposit inside or on top of the membrane pores. Recently, all of these models have been combined and applied towards microfiltration and ultrafiltration of biological fluids [18]. The application involved filtration of virus from a solution containing human plasma IgG and BSA. The combined model was shown to fit well both the modes of filtration – constant pressure filtration and constant flux filtration. More recently, an extension of these models has been suggested to incorporate the effect of flow rate on the deposition of particles for constant pressure filtration [19]. The possibility that a greater flow rate would mean a lower residence time and hence lower deposition of particles inside the pores has been highlighted. The model was shown to provide improved fits for filtration of BSA and human IgG. Another model that combines pore blocking and cake filtration as two stages of filtration to describe the stirred cell filtration through microfiltration membranes has also been proposed [20]. This model accounts for the difference in the thickness of the cake across the membrane and provides good fit for the filtration of BSA. It has, however, not been tested for filtration of viruses.

In this paper, we evaluate previously published data [21] and propose a model that explains the fouling mechanism during viral filtration, wherein viral removal is based entirely on size exclusion as opposed to adsorption [22]. The proposed model accurately fits change in LRV and flux as a function of time for the various different kinds of viral filters that are commercially available. The model has been validated using data on filtration of parvovirus (Φ X-174) by direct-flow small-virus retentive filters.

2. Materials and methods

2.1. Model protein

The model TP (therapeutic protein), a human IgG, is an in-process intermediate from a proprietary protein product from Eli Lilly and Co. Monomer TP was generated by filtration sequentially through two 10.16 cm (4 in.) Viresolve NFP cartridges containing V180 membrane. The detailed procedure for preparing stock protein solution has been described elsewhere [22].

2.2. Phage procedures

Φ X-174 and *Escherichia coli* C (A.T.C.C. 13706) were obtained from the Felix d'Herelle Reference Center for Bacterial Viruses (Quebec, Canada). The detail procedure for preparing stock viral particle has been described elsewhere [22].

2.3. Phage-retention experiments

The filtration methodology included collection and assay of fractions taken over the course of filtration ('grab samples'). The test filters were small-scale disc or hollow fiber devices supplied by the four filter manufacturers for use in scale-down validation studies. Filters from Millipore and Sartorius Stedim

were commercially available devices (25 mm syringe filter units). Filters from Pall were coupons in stainless steel housings. In all cases, the filter is composed of multiple layers "fused" together. These were Optiscale-25 devices with Viresolve NFP membrane (3.5 cm² surface area Millipore), Virosart CPV filter discs (5.3 cm² surface area; Sartorius), Ultipor VF grade DV20 filter discs in a reusable stainless-steel housing unit (Pall housing part number FTK200; 11.1 cm² surface area Pall Corp.), and Planova 20N small-scale hollow-fiber devices (10.0 cm² surface area Asahi Kasei). Filtration experiments were performed at ambient temperature and at vendor-recommended pressures of 207 kPa (30 lbf/in.²) for Viresolve NFP, Virosart CPV, and Ultipor DV20 and of 98 kPa (14.2 lbf/in.²) for Planova 20N. Devices were first pre-wetted with test buffer (without protein). The initial two hold-up volumes (containing mostly buffer, not model protein/phage solution) were discarded and not counted towards throughput in l/m². Challenge solutions were processed through a 0.22- μ m-pore-size filter before being added to the pressure vessel. The system was then pressurized and the volume of filtrate measured every 1–5 min for the duration of the test using precision timers and balances (process fluid density is assumed to be 1 g/ml). The initial flow rate of buffer was used to calculate flow rates at specific time or passage volume points. The degree of decay from the initial flow varied with the particular filter and protein/phage combination. Samples were collected directly from the filter devices at the start of the run, and at the target flow reductions or passage volume points. These samples measure the phage-retention capability of the filter at the time of sample collection, as opposed to pooled filtrate samples typically collected in validation studies, which measure virus retention of the filter over the course of an entire run. LRVs were calculated as follows:

$$\text{LRV} = \log(C_b/C_p) \quad (1)$$

where, C_b is the phage titer in the challenge solution and C_p is phage titer in the filtrate grab samples (permeate).

3. Model

A simple model based on changes in filter resistance and active filter area has been used to model viral filtration. This model has been used to fit the previously published data on viral filtration [21]. Viral particles are modeled as colloidal particles disregarding any specific interactions and considering only passive transport in the system. The model is based on the following arguments:

- Purely steric interaction is assumed to determine the distribution coefficient (Φ), i.e., ratio of concentration of viral particles inside the pore to concentration in solution just at the pore mouth. Using either geometrical [23] or statistical arguments [24], Φ can be related to the ratio of size of viral particles to size of pore of membrane ($\lambda = r_v/r_p$) as given below

$$\Phi = (1 - \lambda)^2 \quad (2)$$

- Viral particles rejected by the membrane form a layer of high concentration called the concentration polarization (CP) layer at the membrane surface which offers additional resistance to filtration.
- The CP layer grows with time as an unsteady-state penetration layer where the growth rate is independent of concentration and depends on diffusion coefficient. Concentration and rejection determine the range of concentrations encountered in diffusion boundary layer. The time dependent thickness of the CP layer (δ) can be related to the time of filtration as given

below [25]

$$\delta = at^b \quad (3)$$

where, a and b are constants. According to penetration layer model, value of b is 0.5. However, due to the interactions between the virus particles and the membrane, the value of b can change with virus concentration and filter properties.

- Assuming a pseudo-steady state where the amount of viral particles brought in by convective flow is equal to the amount diffusing out of the CP layer, the following expression for concentration profile in the CP layer can be proposed

$$C = C_b \exp\left(\frac{J_v x}{D}\right) \quad (4)$$

where, C_b is the bulk concentration, J_v is the permeate volume flux, D is apparent diffusion coefficient of virus particle and x is the position from the bulk. The concentration at the membrane surface (C_s) can be obtained by taking $x = \delta$.

- Some of the viral particles block the pores of the membrane, thereby reducing the available area of membrane for filtration. The rate of reduction in area can be assumed to be proportional to the amount of viral particles transported from bulk towards membrane surface and is given by the following equation [26]:

$$\frac{dA}{dt} = -\alpha J_v C \quad (5)$$

where, α is the factor that relates instantaneous membrane area blocked by the viral particles to the amount of viral particles reaching the membrane surface. It depends on both the nature of the virus particle as well as that of the membrane.

For membrane flux, Darcy's law can be used:

$$Q = \frac{PA}{\mu R_T} \quad (6)$$

where, Q is filtration rate (m^3/s), P is the pressure applied, μ is solution viscosity, R_T is filtration resistance and A is the membrane area available for filtration. R_T consists of membrane resistance (R_m) and the resistance offered by the CP layer near the membrane (R_c). During filtration, both R_c and A change. R_m stands as area specific resistance and is independent of size of the membrane. Specific resistance of the CP layer depends on the porosity of the layer in a way similar to the Carman–Cozeny equation for spherical particles [27]. The average porosity of the layer depends on the average concentration of the viral particles in the layer and so the value of R_c can be calculated by the following equation.

$$R_c = k\langle C \rangle^n \quad (7)$$

where, k and n are constants and $\langle C \rangle$ is the average concentration of the CP layer. Parameters k and n are both independent of membrane and depend only on virus properties. Eq. (7) describes dependence of hydrodynamic resistance of boundary layer on its average concentration and therefore depends on concentration profile of the boundary layer. It is different from Eq. (3) which describes growth of the boundary layer.

Hindered transport inside the pores is used to calculate the flux of viral particles through the membrane pores. It is assumed that only convective component is significant while the diffusive and electro-migration terms are neglected. The flux of viral particles through membrane pore is given as [28]

$$j_i = K_i J_v C_{mp} \quad (8)$$

where K_i is coefficient accounting for the effect of hindrance due to pore walls and C_{mp} is the concentration of viral particles just inside the pore mouth. As λ is expected to be greater than 0.8, K_i is given

by the following equation:

$$K_i = -6.830 + 19.384\lambda - 12.518\lambda^2 \quad (9)$$

Further, C_{mp} can be obtained from the values of Φ and the concentration of viral particles at the membrane surface (C_s) using the following expression:

$$C_{mp} = \Phi C_s$$

Concentration of viral particles in the permeate (C_p) can be obtained by dividing the flux of viral particles (j_i) with the permeate volume flux (J_v) as shown below

$$C_p = \frac{j_i}{J_v} \quad (10)$$

Simultaneous solution of Eqs. (3)–(8) allows us to calculate permeate flux and concentration of viral particles in permeate. In the model, Eq. (4) considers diffusive flux just upstream of membrane while Eq. (8) neglects diffusive flux inside membrane. Eq. (4) describes the situation in upstream of membrane where convective transport of virus towards membrane is balanced by diffusive transport in opposite direction. In view of very low diffusivity of viral particles, a very high concentration gradient is required. Such high concentration gradient is possible because concentration at membrane surface is expected to be very high due to rejection of viral particles by the membrane. Eq. (8) describes the situation inside membrane where diffusive flux is in same direction as convection and concentration gradient is not significant, so the diffusive flux is neglected. Eqs. (3)–(8) describe increase in viral particles' concentration at membrane surface with time of filtration. The concentration depends on flux which depends on both the active area of the membrane and resistance of the boundary layer. Increase in concentration at membrane surface results in decrease in LRV as the partitioning of viral particles at pore mouth remains constant. The model involves use of six unknown parameters (a , b , r_p , α , k and n).

4. Results

Using the previously generated data on viral filtration [21], a general model was built to fit the data for four commercially available filters. This model was used to calculate flux and LRV as a function of time and the load titer of virus. High-titer ($> 10^{13}$ pfu/ml) CsCl-gradient-purified ϕ X-174 was spiked into 2.5 mg/ml TP at 10^7 – 10^{10} pfu/ml and used in filtration experiments. The properties of all the four filters used in this study have been summarized in Table 1. For each filter, three different levels of virus titers were used as a challenge. The filters were characterized on the basis of flux decline and change in LRV during the course of filter operation. Since the same viral particle was used for all modeling studies, values of parameter n ($=0.5$), k ($=1500$) and r_v ($=20$ nm) were kept constant throughout the calculations. To account for filter to filter variations, values of R_m , α and b were varied and for change in viral titer, the parameter ' a ' was adjusted so as to minimize the sum of square of residuals (SSR) with the residual being the difference between experimental data and model prediction.

4.1. Planova 20N

Planova 20N filters generally experienced only modest flux decay in comparison to others filters. The phage passage and flow decay was evident in filters challenged with higher titer of phage ($> 10^9$ pfu/mL). At highest titer of viral particles, a sigma curve

Table 1
Characteristics of viral filters used in this study.

Filter	Vendor	Membrane composition	Effective area (cm ²)
Viresolve NFP	Millipore	Three layers of composite Polyvinylidene fluoride (PVDF)	3.5
Ultipor DV20	Pall	Hydrophilic modified PVDF	11.1
Virosart CPV	Sartorius	Double layer 20 nm polyether sulfone (PESU)	5.3
Planova 20N	Asahi-Kasei	Bundle of straw shaped hollow fibers made of cuprammonium regenerated cellulose	10

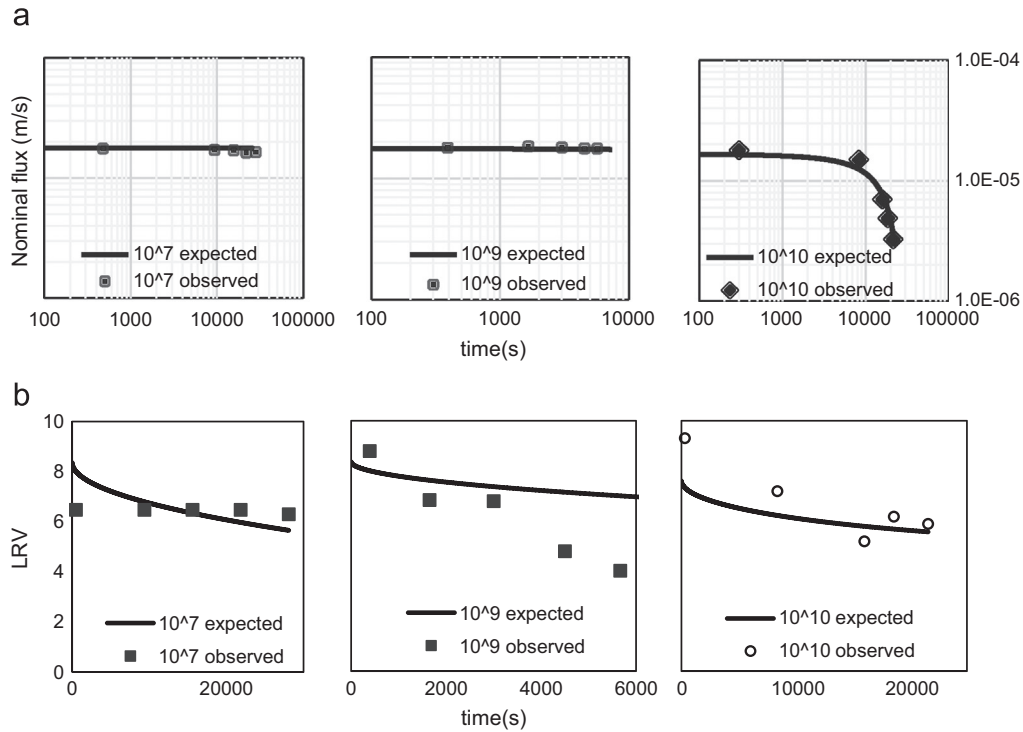


Fig. 1. (a) Nominal flux vs. time profile for the Planova 20N filter at phase titers of 10⁷, 10⁹, and 10¹⁰ pfu/mL. (b) LRV vs. time profile for the Planova 20N filter at phase titers of 10⁷, 10⁹, and 10¹⁰ pfu/mL.

shaped concavity was seen at the end point of filtration (Fig. 1a). In contrast to other filters, the value of λ was in the high range (corresponds to lesser r_p) which signifies higher initial LRV. There was only a modest increase in value of parameter ' a ' from 0.015 to 0.017 and it was constant at 0.017 for higher titers of viral particle. The fraction of pore blocking factor, α , was highest (900×10^{-13}) for this filter. The parameter values and the SSR for flux and LRV are summarized in Table 2. Overall a good fit was obtained for flux but poor fit for LRV for all viral titers used in the proposed model (Fig. 1a and b).

4.2. Ultipor DV20

The flux and LRV profiles were different for this filter in contrast to other filters. The flux was almost constant and only slightly lowered at the end point of operation, though LRV continued to decrease throughout (Fig. 2a and b). The constant resistance R_m in this case was seen to be an order of magnitude higher and had the least pore blocking factor ($\alpha = 1 \times 10^{-13}$) compared to other filters. This may explain nearly flat flux profiles. It is possible that the constant resistance dominated over variable resistance and hence the change in flux was negligible. This is consistent with existing literature on Ultipor filters which states that the flow decay through these filters is very low (until overloaded at concentrations $> 10^{10}$) as compared to the other filters [7,21]. The decay in LRV is significant though. Lower λ (0.966)

implies a larger pore size (r_p) and this is likely to account for the lower LRV values. The parameter defining cake formation, a , increased with increasing phage concentration. The parameter values and the SSR for flux and LRV are summarized in Table 2. Overall a good fit was obtained for both flux and LRV for all viral titers used in the proposed model.

4.3. Viresolve NFP

In contrast to other filters, flux decay was evident even at low phage titers. The flux showed consistent concavity from the start (Fig. 3a). This likely indicates 'protein induced flux decay' (in addition to flux decay due to phage cake formation) as has been described in literature [22]. Due to this effect, the parameter ' b ', which was constant at 0.5 for the previous two filters, had to be changed and the optimal value was found to be 0.35. Increasing titers of purified phage clearly overloaded the filters and thus the parameter ' a ' increased with phage concentration. The flow decline was rapid and more pronounced at higher phage titers which also explains the higher pore blocking factor ($\alpha = 450 \times 10^{-13}$). The parameter values and the SSR for flux and LRV are summarized in Table 2.

While the model provided a fit of medium quality for the LRV profile at modest phage titers (10^8 and 10^9 pfu/mL), it was not able to explain the sharp decline in LRV value when compared to predicted value at the highest titer of 10^{10} pfu/mL (Fig. 3b).

Table 2
Values of the fitted parameters for the different filters and conditions explored in this investigation. The optimal values were obtained via minimizing the sum of square of errors (SSR).

Filter	n	k	$R_m (\times 10^9)$	λ	$\alpha (\times 10^{-13})$	b	pfu/mL	a	SSR LRV	SSR flux
Planova 20N	0.5	1500	5.5	0.980	900	0.5	10^7	0.015	3.14	5.1×10^{-12}
							10^8	0.017	16.06	9.7×10^{-13}
							10^{10}	0.017	6.74	9.2×10^{-12}
Ultipor DV20	0.5	1500	30	0.966	1	0.5	10^7	0.09	1.63	5.3×10^{-13}
							10^8	0.100	2.27	6.11×10^{-12}
							10^9	0.140	1.09	2.23×10^{-13}
Viresolve NFP	0.5	1500	2.0	0.976	450	0.35	10^8	0.0025	3.62	1.87×10^{-10}
							10^9	0.0025	2.86	1.15×10^{-10}
							10^{10}	0.0030	18.77	1.63×10^{-09}
Virosart CPV	0.5	1500	5.0	0.998	5	0.42	10^7	0.024	19.68	1.12×10^{-10}
							10^8	0.018	12.35	1.11×10^{-10}
							10^9	0.024	5.18	3.34×10^{-11}

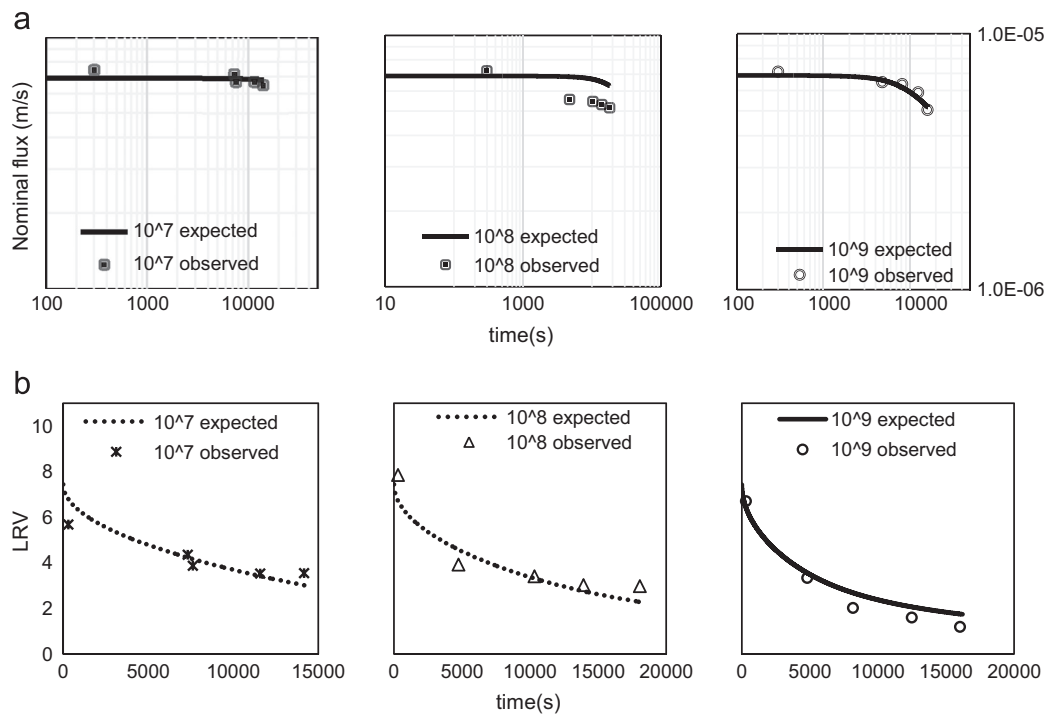


Fig. 2. (a) Flux vs. time profile for the Ultipor DV20 filter at phase titers of 10^7 , 10^8 , and 10^9 pfu/mL. (b) LRV vs. time profile for the Ultipor DV20 filter at phase titers of 10^7 , 10^8 , and 10^9 pfu/mL.

4.4. Virosart CPV

As with the case of Viresolve NFP filters, this filter also exhibited flux decline relative to buffer flux even at low phage concentration (Fig. 4a). This flux decline also appeared to be protein related and phage titer only had modest effect. Thus model parameter ' b ' was calculated to be 0.42 (less than 0.5) in contrast to that for Ultipor DV20 and Planova 20N filters where flux decline was attributed only to phage cake formation. The LRV decline with the processing time was relatively less which explains the higher λ ($=0.998$) and the lower r_p obtained for this filter (Fig. 4b). The value of model parameter ' a ' changed with increase in titer of phage particle. This is likely due to the higher filter-to-filter variability in Virosart filters which has also been highlighted in the literature [29,30]. This filter is known to exhibit significant filter to filter variability in LRV as well as LRV decline, likely due to the membrane composition. The parameter values and the SSR for flux and LRV are summarized in Table 2. The model was able to

explain all the observed values of LRV and flux with a fit of medium quality.

5. Discussion

In the model, the concentration of viral particles in boundary layer is allowed to increase to an extent that diffusive flux due to concentration gradient balances the convective flux. However, the maximum concentration (at the membrane surface) of viral particles is limited by a critical volume fraction value beyond which precipitation will occur. Among all filters and concentrations used in this work, the highest viral concentration at (Virosart) membrane surface was 1.43×10^{10} pfu/ml and corresponds to a volume fraction of 0.0048. Thus the largest volume fraction of viral particles is too small to result in precipitation. However, there may exist cases where volume highest concentration in boundary layer does not represent realistic values and the model needs

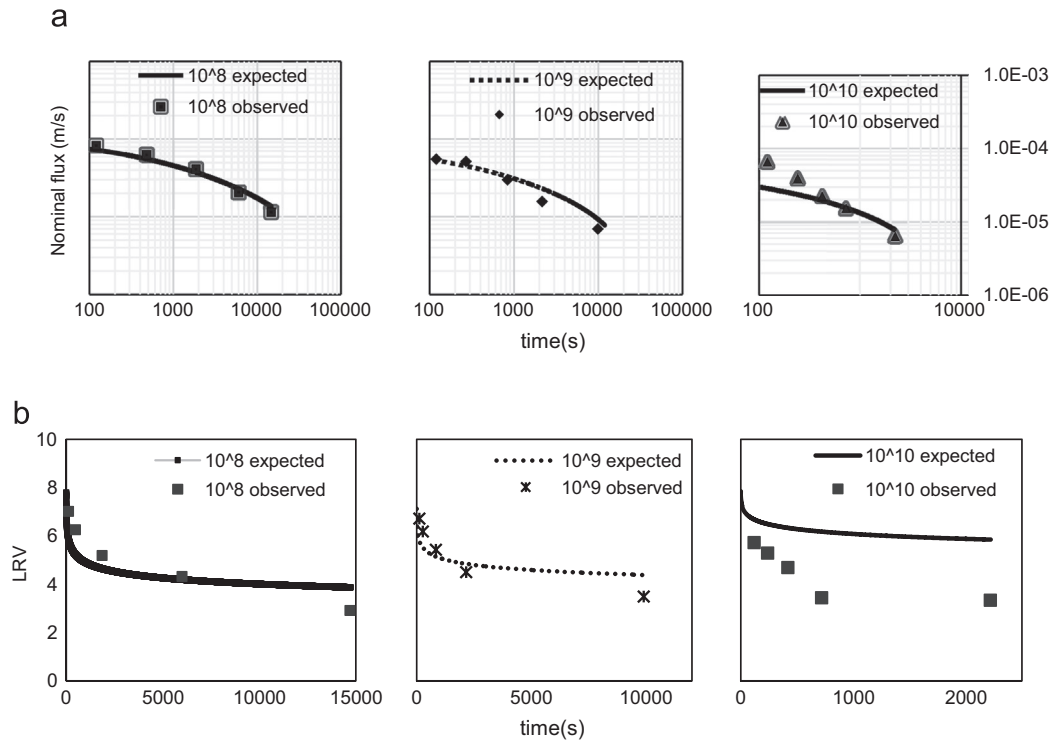


Fig. 3. (a) Flux vs. time profile for the Viresolve NFP filter at phase titers of 10^8 , 10^9 , and 10^{10} pfu/mL. (b) LRV vs. time profile for the Viresolve NFP filter at phase titers of 10^8 , 10^9 , and 10^{10} pfu/mL.

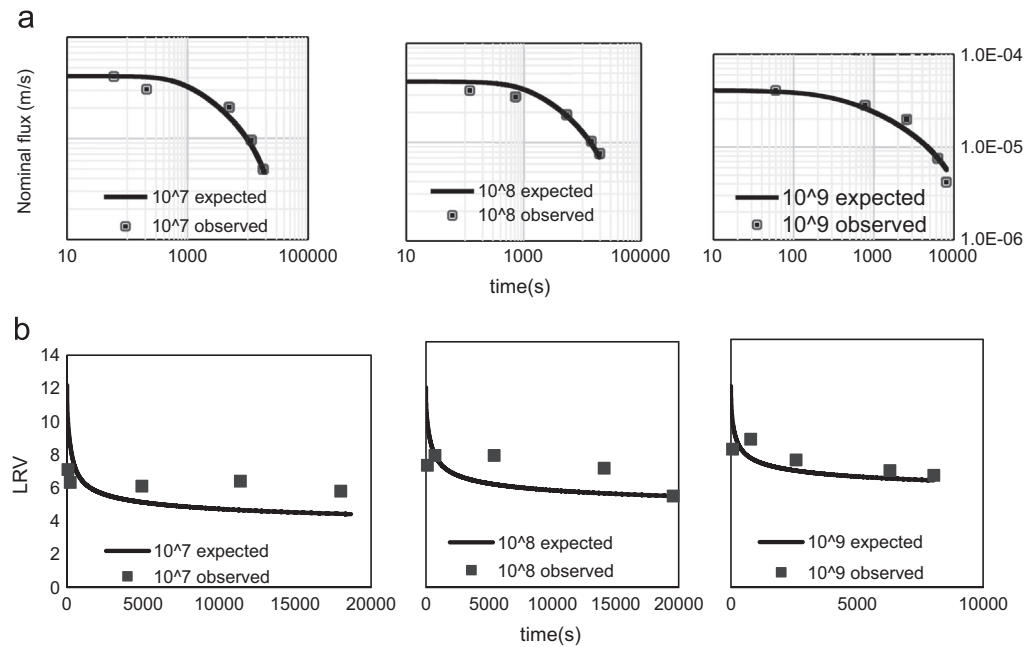


Fig. 4. (a) Flux vs. time profile for the Viosart CPV filter at phase titers of 10^7 , 10^8 , and 10^9 pfu/mL. (b) LRV vs. time profile for the Viosart CPV filter at phase titers of 10^7 , 10^8 , and 10^9 pfu/mL.

modification to account for formation of cake layer of viral particles.

Model does not explicitly account for the details of membrane structure like its asymmetric nature. Viral particles accumulate inside membrane structure if the pores (of support layer) on upstream side are larger than viral particles. Accumulation of viral particles inside membrane would lead to additional resistance. Parameter α describes the change in active area of the membrane due to pores blocked by viral particles. However, best-fit values of α (Table 2) for the membranes which trap viral particles were

much higher than for other membranes. Thus, α also takes into account the increased resistance of membrane due to internal-accumulation of viral particles but in terms of change in active area. It is also possible to enhance the model by including separately the resistance due to constriction of pores on accumulation of viral particles, but that will increase the number of parameters of model further.

As is evident from Figs. 1–4, the proposed model is successful in fitting observed flux and LRV for viral filters from several vendors. It explains both the phage and protein dominated flux decline for

the range of titers used. Table 2 displays the parametric constants for the different filters evaluated in this study. The flat flux profile obtained for Ultipor DV20 is explained by the high constant resistance R_m ($\sim 10^{10}$) obtained for this filter. Other filters exhibited similar R_m values ($\sim 10^9$). Different LRV observed with the different filters can be attributed to differences in ratio of size of viral particles to size of pore of membrane $\lambda (=r_v/r_p)$. Since same viral particles ϕ X-174 was used r_v was constant, while radius of pore r_p (average) varied. The highest value of λ (0.998) for Virosart CPV correlates to higher LRV while the lowest value (0.966) was for Ultipor DV20 resulting in the lowest LRV. Flux decline due to fouling by viral particles was correlated to the pore blocking factor (α). This was clearly evident for Planova 20N where a sharp decline in flux was observed at higher phage concentration. Flux decline and LRV due to cake formation correlated to a and b. The value of parameter 'b' was 0.5 when decline was largely due to phage as with Planova 20N and Ultipor DV20. When protein dominated cake formation was expected then the value of 'b' was lower as with Virosart CPV and Viresolve NFP. In general, value of parameter 'a' increased with increasing titer of phage.

The LRV predicted by the model for Virosart CPV at short times (where experimental data cannot be taken) are high. At other times, the model trend adequately matches the flat-nature of experimental data. It may be due to presence of larger pores in the filter that lower the LRV initially. For Planova 20N also the discrepancy indicates that some pores of larger size result in reduction in LRV at low concentration, but get blocked earlier at higher virus concentrations and result in a declining LRV. Resistance due to protein can be the more likely reason for this but there could be other reasons as well. For example, roughness of filter could also affect how concentration develops near membrane surface. While it would be of interest to study filtration without protein, such an experiment may not provide significant general information because resistance of the boundary layer could strongly depend on protein virus pairing (due to shape or other long range interactions). The model in this work shows that even in absence of such details, the LRV behavior could be explained from very well to a satisfactory level for most of the filters that were examined in this study.

6. Conclusions

A model has been proposed for viral filtration. The model accounts for resistance to filtration due to deposition of particles inside the membrane and deposition as a concentration polarization layer. The change in filter area due to both of these factors has also been considered. The model has been examined for four different filters. The quality of the fit was found to vary from filter to filter, perhaps due to differences in filter design as well as variability in filter performance. Each of the filters shows a different trend of flux decline and LRV. The six parameters in the model can be adjusted according to the filter used and virus concentration. This ensures that the model can successfully explain the seemingly different trends that are seen experimentally. Time profile of flux and LRV are well explained by the model for most cases.

Nomenclature

LRV	log reduction value
ϕ	distribution coefficient
J_v	permeate volume flux
j_i	viral particles flux

D	diffusion/dispersion coefficient
Q	filtration rate
A	membrane area available
R_T	filtration resistance
R_m	membrane resistance
R_c	resistance offered by the high viral-concentration layer near the membrane
$\langle C \rangle$	average concentration of viral particles
C_{mp}	concentration of viral particles
C_p	permeate concentration
C_b	bulk concentration
C_s	concentration of viral particles at the membrane surface
pfu	plaque forming units

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